

Block-A-Mole: The Sustainability Frontier of Moving-Target Censorship Resistance

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Abstract—Internet censorship affects over four billion people, and every deployed circumvention system shares one structural weakness. Its endpoints are fixed and discoverable, so a patient censor eventually enumerates and blocks them. A recent class of systems instead makes infrastructure a *moving target*, rotating endpoints across commercial cloud address space faster than a censor can block them. These systems demonstrate that the idea works, but the field has no theory of *when* it works, and rotation intervals and pool sizes are chosen by intuition. We give the first formal account. We model the censor and defender interaction as a continuous-time timing game on a combinatorial (address, domain) space, generalizing FlipIt to a collateral-bounded adversary, and we prove a *sustainability frontier* that separates the configurations a censor can defeat from those it cannot. Our central finding is that under the Great Firewall’s 2024 shift to blocking QUIC and TLS *by domain*, raw rotation speed is not the binding constraint. Whether rotation wins is governed by a single dimensionless quantity, the *domain burn rate* $\beta = \lambda_{\text{disc}}/\lambda_{\text{intro}}$, the ratio of how fast the censor blocks the defender’s domains to how fast the defender mints fresh ones. We derive a closed-form availability law, prove that address rotation alone ($\beta > 1$) cannot sustain high availability regardless of how fast endpoints rotate, and characterize the sustainability frontier β^* . We confirm every analytical result in an open, model-level censor–defender simulator that needs no privileged access or cloud deployment, reproducing the predicted phase transition at β^* under adversary profiles representative of the GFW, Russia’s TSPU, and Iran, and we show the frontier is robust to state-dependent discovery and bursty, provider-correlated burns. The validation is model-level, confirming that the analysis and an independent simulation agree rather than fitting a specific censor. The theory replaces the common design heuristic of rotating faster with a precise operating condition, namely keeping the domain economy ahead of the censor, and supplies an open, reproducible testbed for the censorship-resistance community.

1. Introduction

Internet censorship is among the defining challenges of the digital age. Freedom House reports that global Internet freedom has declined for fourteen consecutive years, and over 60 governments routinely block websites, throttle connections, or shut down networks, affecting an estimated four billion people [1]. The sophistication of this control

has escalated and is now worldwide. China’s Great Firewall (GFW) performs real-time protocol fingerprinting, active probing, and passive detection of fully encrypted traffic [2], [3], and in April 2024 it began decrypting QUIC Initial packets at scale and blocking connections *by domain* [4], reaching the very transport that modern moving-target systems rely on for session continuity. Russia’s TSPU enforces ISP-level protocol blocking at line speed [5], and Iran runs a national filtering infrastructure able to disconnect the country during unrest [6], [7].

Across two decades of circumvention research, spanning Tor and its pluggable transports [8], [9], domain fronting [10], Snowflake [11], and refraction networking [12], one structural weakness recurs. *Any system with fixed, discoverable endpoints is eventually enumerated and blocked.* Tor relay IPs are public, bridges are harvested from every distribution channel [13], and pluggable transports are fingerprinted within weeks [2]. This is a consequence of *static infrastructure* rather than an implementation defect.

A recent line of work inverts this asymmetry by making infrastructure a *moving target*, rotating endpoints across commercial cloud address space faster than a censor can act on what it discovers. SpotProxy [14] runs circumvention proxies on cloud spot instances, exploiting the providers’ own reclamation churn to rotate IP addresses at low cost. NetShuffle [15] shuffles proxy addresses through programmable switches in cooperating edge networks, and CensorLess [16] places bridges inside short-lived serverless functions. In the DNS setting, the NinjaDoH system [17] rotates DNS-over-HTTPS resolvers across ephemeral hyperscaler instances whose IP addresses change continuously, announced through a decentralized naming layer [18], and reports high availability at a cost of cents per day. What unifies these designs is *collateral freedom*. A censor that blocklists a hyperscaler’s IP range to suppress one moving-target service also blocks the legitimate workloads its economy depends on. AWS alone operates more than 10^8 public IPv4 addresses, behind which sit domestic banks, retailers, and government services, so severing that range to chase one rotating endpoint imposes a domestic cost the censor will not accept. Rotation thus makes the censor’s discovery work stale faster than it can act on it, at least for the address.

These systems prove the idea works, but the field has *no theory of when* it works. Every system picks rotation intervals and pool sizes by intuition, and three first-order questions have no principled answer.

- 1) *When does rotation provably win?* There is no characterization of the conditions on rotation rate, pool size, collateral budget, and the rate at which a censor burns the defender’s domains under which availability is *guaranteed*, nor of the censor’s optimal counter-strategy.
- 2) *What is the binding resource?* Systems rotate IP addresses, but the GFW now blocks by domain. Whether the scarce resource is the address or the domain changes the entire design calculus, and has never been settled.
- 3) *Is “rotate faster” even the right lever?* The recurring intuition that a defender should out-rotate the censor has never been made precise, and, as we show, is misleading under a domain-filtering censor.

This paper answers all three.

Contributions. This paper supplies the missing theory and an open instrument to test it. We make the following contributions.

- **A timing-game model of moving-target censorship resistance** (§4). We formalize censor–defender interaction as a continuous-time timing game on a combinatorial (address, domain) space, generalizing the FlipIt model of stealthy takeover [19] and Stackelberg models of moving target defense [20] to a setting with a *combinatorial resource* and a *collateral-bounded* adversary (Definitions 1–2).
- **The address resource is free, the domain resource is binding** (§5–6). We prove that address blocking incurs a denial probability that decays geometrically in the number of endpoints (Lemma 1), so address rotation defeats an address-only censor. The scarce resource is the *domain* pool, which we model as a birth–death process (Proposition 1).
- **A closed-form availability law and an impossibility result** (§6.1). We derive availability in closed form (Theorem 1) and prove that whenever the domain burn rate $\beta = \lambda_{\text{disc}}/\lambda_{\text{intro}} > 1$, availability is bounded away from one *regardless of rotation speed* μ , pool buffer, or endpoint count (Theorem 2). Sustained availability requires $\beta < \beta^* \leq 1$, and we characterize the frontier β^* (Theorem 3) and the censor’s optimal Stackelberg response (Proposition 3).
- **An open censor–defender simulator** (§7). We release an event-driven simulator that instantiates the game, exercised here with adversary profiles representative of documented GFW, TSPU, and Iranian behavior and designed to be calibrated to public measurement archives. It needs no privileged access or cloud deployment, giving the community a reproducible, open testbed.
- **Validation through a phase transition at β^*** (§8). The simulator reproduces every theoretical prediction, including a sharp phase transition in availability at β^* , and shows that increasing rotation speed does not move the frontier. We convert the theory into a concrete domain-economy operating recipe.

The result reframes a decade of system design. The governing parameter is the domain-introduction rate, not the rotation interval.

2. Background and Related Work

Censorship resistance is rich in *systems* but poor in *theory*, and two systematizations survey the landscape and note the same gap [21], [22]. We review the lineage thematically and make the gap explicit.

2.1. The Static-Infrastructure Arms Race

Two decades of circumvention have followed one pattern. A transport is deployed, the censor learns to detect or block it, and a successor appears. Tor [8] and its pluggable-transport framework [9] anchor the lineage. Early designs tunneled through allowed services (Infranet [23]) or imitated allowed protocols such as SkypeMorph [24], format-transforming encryption [25], and the programmable obfuscator Marionette [26]. Houmansadr et al. showed that such *mimicry* is fundamentally flawed because perfect behavioral imitation is infeasible [27]. Randomized transports (ScrambleSuit [28], obfs4 [29]) make traffic look like nothing, yet are themselves fingerprinted [30], and “fully encrypted” protocols are now detected wholesale by the GFW [2]. Probe-resistant proxies that present genuine TLS, such as REALITY [31] and Shadowsocks [32], resist active probing but are still detectable [33] and rely on *static* endpoints. Tunnel-in-carrier designs (FreeWave [34], Protozoa [35], Balboa [36]) embed data in innocuous flows but inherit the same structural weakness. Any fixed, discoverable endpoint is eventually enumerated and blocked [13], [37]. This work assumes the censor finds every address and rotates before blocking takes effect.

2.2. Refraction and Decoy Routing

A different escape from static endpoints places the circumvention proxy *inside* the network, at a cooperating ISP, so there is no discrete address to block. Decoy routing [38] and its instantiations Telex [39], Cirripede [40], and TapDance [41] realize this, with later work resisting traffic analysis (Slitheen [42]), routing-around-decoy attacks (Waterfall [43]), reusing unused address space (Conjure [12]), and pushing toward real deployment [44]. Refraction is powerful but requires *ISP cooperation*. Our setting assumes no such ally and instead exploits the economics of commodity clouds.

2.3. Moving-Target Circumvention

The line closest to ours makes infrastructure a moving target on commodity clouds. SpotProxy [14] rotates proxy IPs using spot-instance reclamation churn. NetShuffle [15] shuffles proxy addresses through programmable switches at cooperating edges, CensorLess [16] places bridges in serverless functions, and in the DNS setting NinjaDoH [17] rotates DoH resolvers across hyperscaler instances. Domain fronting [10] and its decline under provider pressure [45] are the cautionary precursor, and Turbo Tunnel [46] supplies the session persistence rotation needs. Deployed tools

(Psiphon [47], Lantern [48], Snowflake [11]) reach millions but are likewise point designs. All are validated empirically under specific conditions and, most relevant here, come with *no formal guarantee* about when rotation defeats a censor. We supply that guarantee and a theory that applies across these designs.

2.4. Discovery and Anti-Enumeration

Rotation is of little use if users cannot find the current endpoints, and any directory a user can query a censor can query too. Bridge distribution has accordingly evolved from user-generated-content channels (Collage [49]) and measurement-driven dissemination (Proximax [50]) to reputation- and social-graph-protecting schemes (rBridge [51], Salmon [52], Lox [53]), all fighting systematic harvesting [13]. Decentralized naming (IPFS/IPNS [18], ENS [54], Handshake [55]) offers censorship-resistant lookup but is publicly queryable, and the cryptographic tools for a private directory (anonymous credentials [56], private information retrieval [57]) have not been combined with rotation. Discovery is orthogonal to the present paper’s reachability analysis, and we note it because a complete system needs both.

2.5. Censor Capabilities and Measurement

Our threat model is grounded in measured censor behavior. The GFW has been characterized blocking Tor [58], actively probing hidden servers [37], injecting DNS responses with wide collateral damage [59], [60], detecting Shadowsocks [3] and fully encrypted traffic [2], and, decisively for this work, filtering QUIC and TLS *by domain* [4], with GFWatch documenting hundreds of thousands of persistently blocked domains [61]. Other censors are comparably documented, including Russia’s TSPU and decentralized control [5], [62], Iran’s filtering and protocol whitelisting [6], [7], and country-scale shutdowns [1], [63], [64]. A mature measurement ecosystem makes this behavior observable, including OONI [65], Censored Planet [66], ICLab [67], Quack [68], Augur [69], Iris [70], Satellite [71], Encore [72], and global filter mapping [73]. Our simulator draws its adversary mechanisms from these documented sources and is designed to be calibrated to the public archives, neither of which requires privileged access.

2.6. Encrypted DNS, SNI, and ECH

Because the censor now filters by name, name-hiding is directly relevant. Encrypted DNS (DoH/DoT) and oblivious variants [74] reduce metadata leakage, and ESNI/ECH aim to hide the server name in TLS [75], [76], but measurement shows these are themselves blocked or degraded under censorship [77], [78], and the GFW’s 2024 QUIC-by-domain filtering [4] reaches the very transport moving-target systems rely on. Name encryption raises the censor’s cost but does not remove the domain as a scarce, blockable resource, which is the premise our theory formalizes.

2.7. Traffic Analysis and Its Defenses

A complementary threat is per-flow detection. Deep-learning website-fingerprinting attacks [79], [80], [81], [82], [83], [84], portable few-shot variants [85], pretrained transformer classifiers [86], and compressive analysis [87] keep raising the bar, answered by padding, regularization, and traffic-splitting defenses [88], [89], [90], [91], [92], [93] and adversarial perturbations [94], [95]. Bandwidth lower bounds are known for *anonymity* (the anonymity trilemma [96]) but not for moving-target availability. We do not address per-flow detectability, and our reachability results compose with any of these transports.

2.8. Game-Theoretic Moving Target Defense

The timing-game machinery we build on comes from a literature that has not met censorship. FlipIt models stealthy takeover as a timing game [19]. Stackelberg and broader formulations derive optimal MTD timing and address mutation against rational attackers [20], [97], [98], [99], [100], [101], [102], and security games provide the equilibrium and online-learning foundations [103], [104], [105], with adversarial bandits and partial monitoring [106], [107] the natural tools for an adaptive-censor controller. These works target enterprise defense and never model a combinatorial address space, a collateral-bounded adversary, or domain-based filtering, while censorship moving-target systems, which run on transient cloud servers [108], borrow none of the game-theoretic machinery. We bridge them.

2.9. The Theory Gap

Across moving-target censorship systems the literature offers measured performance but no theory. There is no characterization of when rotation wins, no condition separating the configurations a censor can defeat from those it cannot, and no account of the censor’s optimal counter-strategy. The closest prior theory, game-theoretic moving target defense, proves guarantees only for enterprise settings and never models a combinatorial address space, a collateral-bounded adversary, or domain-based filtering. This paper closes that gap for the rotation layer, supplying the missing conditions as closed-form results together with an open simulator any group can rerun to reproduce and build on them.

3. Threat Model

We model a nation-state censor consistent with documented capabilities of the GFW, Russia’s TSPU, and Iran’s infrastructure [2], [4], [5], [7].

Position and detection. The censor controls choke points (gateways, ISP middleboxes) and can observe, delay, modify, inject, or drop packets. It runs deep packet inspection that classifies protocols, extracts metadata (SNI, DNS, certificate fields), and inspects QUIC Initial packets to

filter by domain [4]. It computes flow-level statistics and actively probes suspected endpoints [37]. Having observed an endpoint, the censor may block its *address* and/or its *domain*, and blocks persist. A censor that later unblocks only helps the defender, so persistent blocking is the conservative assumption.

Economic constraint (collateral freedom). The censor must preserve economic utility. Blocking a hyperscaler IP range to suppress one service also blocks the legitimate workloads co-located there. We model this as a bounded *collateral budget* γ , the maximum fraction of a provider’s address space the censor will block before incurring unacceptable domestic cost. γ erodes where a domestic cloud ecosystem reduces foreign dependence (notably China). Domains enjoy no such collateral protection, since blocking a defender’s registrable domain costs the censor essentially nothing.

Provider cooperation is a correlated burn, not a separate failure mode. Providers also act on the defender directly, through account suspensions, abuse takedowns, payment controls, and rate limits. We do not treat this as out of band. A suspension empties that provider’s pool at once, which is exactly the correlated, provider-scoped takedown analyzed in §8.7 and answered by diversifying across independent providers (§8.8). The one regime that dissolves the premise is *universal* provider collusion, where every host de-platforms the service and collateral freedom fails entirely (the failure that killed domain fronting [45]); that is a boundary condition on the moving-target approach itself, outside any rotation policy, and we do not claim to defend it.

Adaptive adversary. We allow the censor to be adaptive, so it may revise its classifier and its block/probe policy over time. We do not, however, require it to be omniscient. It discovers endpoints and domains through the observable channels above, at rates λ_a and λ_{disc} that summarize its discovery throughput. A more capable censor is modeled by larger rates, and we report results as functions of these rates rather than at a single operating point. Where the censor’s true rates are uncertain, we configure them *adversary-favorably*, so the guarantees we report are conservative.

Dynamics of the collateral budget. γ is not a constant of nature but a policy choice that shifts with the domestic economy. It is large where foreign hyperscalers are economically load bearing and erodes as a domestic cloud ecosystem grows (China’s, and Russian import substitution). Our analysis treats γ as a parameter and reports guarantees as functions of it. The qualitative conclusions hold across the plausible range, and we show in §5 that for the address layer *any* $\gamma < 1$ leaves rotation the winner, so the action is entirely on the (collateral-free) domain layer.

Out of scope. We do not provide sender anonymity, and a user’s ISP sees connections to cloud IPs. Users needing both can layer Tor over a moving-target entry point. We assume the user can reach some cloud address initially

(the bootstrap and discovery problem is orthogonal and is treated by anti-enumeration directories, outside this paper’s scope). We also do not model per-flow traffic detectability. Our concern is reachability under blocking, which composes with any probe- or analysis-resistant transport.

4. The Rotation Game

We formalize the recurring intuition that a defender should “rotate faster than the censor blocks” as a continuous-time timing game.

Definition 1 (Moving-target censorship game). Let $\mathcal{A}$ be the address space ($|\mathcal{A}| = N$, the IP addresses an endpoint can be reached at) and $\mathcal{D}$ the domain space ($|\mathcal{D}| = M$, the registrable names it is reached by). The *defender* maintains n active endpoints. Each endpoint independently refreshes its address (and sub-domain) at rate μ (mean interval $\Delta t = 1/\mu$), and the defender mints fresh registrable domains at rate λ_{intro} . The *censor* observes each active endpoint through discovery (active probing, traffic correlation) at rate λ_a per endpoint and discovers/blocks registrable domains at aggregate rate λ_{disc} . On discovery the censor may add the address to a blocklist $\mathcal{B}_{\text{ip}} \subseteq \mathcal{A}$ (subject to the collateral constraint $|\mathcal{B}_{\text{ip}}| \leq \gamma N$) and/or the domain to $\mathcal{B}_{\text{dom}} \subseteq \mathcal{D}$. Blocks persist. An endpoint is *reachable* at time t if neither its address nor its domain is blocked. The defender’s payoff is the time-average fraction of t at which ≥ 1 endpoint is reachable.

Definition 2 ((α, T) -availability). A configuration is (α, T) -available if, against the censor’s best response, the probability that at least one endpoint is reachable throughout any interval of length T is at least α .

Two observations sharpen the problem and depart from prior framings. First, because N is effectively unbounded ($> 10^8$) and addresses are cheap, address exhaustion is not the binding constraint, a fact prior systems exploit implicitly but never state. Second, since the GFW now filters QUIC and TLS by domain [4] and domain blocklisting is the dominant persistent mode [61], the scarce, costly resource is the *domain* pool. Domains require registration and certificate issuance, and once burned are blocked persistently. This motivates the central quantity of this paper, the *domain burn rate*

$$\beta = \frac{\lambda_{\text{disc}}}{\lambda_{\text{intro}}}, \quad (1)$$

the dimensionless ratio of the rate at which the censor discovers and blocks defender domains to the rate at which the defender introduces fresh ones.

Two resources, two timescales. The game decomposes into a *fast* address layer and a *slow* domain layer. On the fast layer, each endpoint’s address races discovery (λ_a) against rotation (μ). On the slow layer, the defender’s registrable-domain economy races minting (λ_{intro}) against burning (λ_{disc}). A modern defender can rotate sub-domains and IPs cheaply (free-tier platform sub-domains co-located

with millions of legitimate services, infrastructure-as-code provisioning [109]), but a fresh *registrable* domain costs money and certificate issuance. The remainder of the paper shows that the slow layer binds.

Modeling choices and assumptions. Three choices make the game tractable while keeping it faithful. (i) We take discovery and minting to be Poisson, which makes the layers Markov. Appendix B removes this assumption and shows the geometric bound is distribution free, while heavy-tailed (occasionally very slow) discovery only helps the defender. (ii) We treat the n endpoints as exchangeable and independent on the address layer, which is exact when rotations draw fresh addresses independently, as the design intends. (iii) We let a burned registrable domain be replaced by a *sub-domain* switch onto any other live domain, an essentially free operation, so the costly event is minting a fresh registrable domain rather than migrating an endpoint. This separation of cheap rotation from costly minting is what makes β , and not μ , the governing quantity, and it mirrors how deployed systems actually use wildcard and free-tier platform sub-domains. We relax the independence of burns (correlated registrar takedowns) in §8.7.

A worked micro-example. Suppose a defender runs $n = 8$ endpoints, rotates four times faster than discovery ($\mu/\lambda_a = 4$), holds a buffer of $k_{\max} = 8$, and faces $\beta = 0.8$. The address layer is essentially perfect, since all eight addresses are simultaneously blocked with probability $(1/5)^8 \approx 2.6 \times 10^{-6}$, and the domain pool sits near full, empty only about 4% of the time, so the system is reachable about 96% of the time *provided* the defender keeps minting. Double the burn rate to $\beta = 1.6$: nothing about rotation changes, yet the pool drifts toward empty and time-average availability falls to about 60% (lower still over a sustained session). The entire difference is the domain economy, which we now make precise.

4.1. Relation to FlipIt and the Combinatorial Resource

Our game generalizes FlipIt [19] in three ways that matter for censorship. FlipIt contests a *single* shared resource whose control alternates; ours is *combinatorial*, an (address, domain) pair with $|\mathcal{A}|$ effectively unbounded, so the defender never re-contests a burned point and simply moves to a fresh one. FlipIt’s moves are symmetric, undoable “flips”; here the censor’s block is *persistent* (§3) while rotation is cheap, an asymmetry that favors the mover with the larger resource. And we add a *collateral budget* γ capping how much address space the censor may seize, with no FlipIt analogue. Classical FlipIt is the degenerate case of a single non-replenishable domain ($M = 1$, $\lambda_{\text{intro}} = 0$), where the pool is a pure death process and availability collapses regardless of rotation speed; with a *replenishable*, *collateral-protected*, *combinatorial* resource the conclusion inverts on the address layer and is governed, on the domain layer, by the single ratio β .

5. The Address Resource Is Free

Here “free” means abundant and cheap to replace rather than literally zero-cost. On a commodity cloud a defender can provision a fresh instance with a new public IP in seconds and at negligible marginal cost, drawing from a provider’s pool of $> 10^8$ addresses. This is the sense in which the address resource is free, and it is *not* true of registrable domains (§6), which is why the two resources behave so differently. We now formalize why address blocking is a losing strategy. Each endpoint’s address is a two-state continuous-time Markov chain, moving from *clear* to *blocked* at the discovery rate λ_a and from *blocked* to *clear* at the rotation rate μ (a rotation moves the endpoint to a fresh address drawn from the unblocked pool, which has size $\geq (1 - \gamma)N$).

Lemma 1 (Geometric address bound). *In steady state, a single endpoint’s address is blocked with probability $p_{\text{ip}} = \lambda_a/(\lambda_a + \mu)$. With n independent endpoints, the instantaneous address-denial probability is*

$$P_{\text{deny}}^{\text{ip}} = \prod_{i=1}^n \Pr[\text{endpoint } i \text{ blocked}] \leq \left(\frac{\lambda_a}{\lambda_a + \mu} \right)^n, \quad (2)$$

which decays geometrically in n . The inequality is strict whenever the collateral cap $|\mathcal{B}_{\text{ip}}| \leq \gamma N$ binds, since the censor cannot block every discovered address.

Proof. The two-state chain has generator with off-diagonal rates λ_a (clear→blocked) and μ (blocked→clear), and its stationary distribution gives $p_{\text{ip}} = \lambda_a/(\lambda_a + \mu)$. Endpoints rotate independently to fresh addresses, so the n block events are independent and the product bound follows. A fresh address is drawn from the unblocked pool of size $\geq (1 - \gamma)N$, and once $|\mathcal{B}_{\text{ip}}|$ reaches γN the censor cannot block newly discovered addresses, which only lowers p_{ip} . $\square$

Corollary 1. *For any target $\varepsilon \in (0, 1)$, address-denial is below ε once $n \geq \ln(1/\varepsilon) / \ln((\lambda_a + \mu)/\lambda_a)$. Even when the censor discovers endpoints faster than they rotate ($\lambda_a \gtrsim \mu$), modest redundancy drives denial to zero.*

Lemma 1 explains why address blocking is a losing strategy for the censor. Addresses are free to the defender and, under the collateral budget γ , the censor cannot even block everything it discovers, let alone keep pace with rotation. The substantive analysis therefore concerns the domain resource.

6. The Domain Resource Is Binding

The blockable unit, and what it costs. The domain resource is the *blockable unit*: the finest name at which the censor can place a persistent block without unacceptable collateral [4], [61]. Which name plays that role depends on where the defender hosts, and three cases recur. (a) A defender-owned registrable domain (example.com):

the censor blocks at the registrable level, so the domain and every sub-domain under it share one fate. It is therefore a single unit, costs a few dollars a year, and its sub-domains supply free SNI variation but burn together. (b) A sub-domain of such a domain is consequently not an independent unit. (c) A free-tier platform sub-domain (*.platform, co-located with millions of tenants): here the censor cannot block the platform suffix without crippling collateral, so it must block the specific sub-domain by SNI. Each such sub-domain is then an independent unit, near-free to mint, that inherits the platform's collateral protection. In every case a fresh unit must be registered or provisioned and given a browser-trusted certificate, and once blocked it is persistently spent, so minting a fresh unit, not switching SNI within a live one, is the costly operation. The rates λ_{intro} and λ_{disc} meter that unit, and a real economy mixes (a) and (c): a few self-controlled registrable domains plus many cheap, collateral-protected platform sub-domains.

The binding dynamics are thus a second timing game on $\mathcal{D}$. The defender holds a buffer of at most k_{max} live, unblocked units, a stockpile knob where more buffer costs more. Let $K(t)$ be the number of live unblocked units. Fresh units are minted at rate λ_{intro} (while $K < k_{\text{max}}$), and the censor burns a live unit at rate λ_{disc} (while $K \geq 1$). Endpoints draw names from the live pool, and when a unit is burned, the endpoints on it migrate to another live unit if one exists, a cheap switch (the costly event is *minting*, not *migrating*).

Proposition 1 (Domain-pool dichotomy). *$K(t)$ is a birth-death chain [110] on $\{0, \dots, k_{\text{max}}\}$ with birth rate λ_{intro} and death rate λ_{disc} . Its stationary distribution is $\pi_k \propto (1/\beta)^k$, giving*

$$\pi_0 = \Pr[K=0] = \begin{cases} \frac{1 - 1/\beta}{1 - (1/\beta)^{k_{\text{max}}+1}}, & \beta \neq 1, \\ \frac{1}{k_{\text{max}} + 1}, & \beta = 1. \end{cases} \quad (3)$$

For $\beta < 1$ the chain drifts toward a full buffer (π_0 is exponentially small in k_{max}), while for $\beta > 1$ it concentrates near 0.

Proof. Detailed balance gives $\pi_k \lambda_{\text{intro}} = \pi_{k+1} \lambda_{\text{disc}}$, so $\pi_{k+1} = \pi_k (\lambda_{\text{intro}} / \lambda_{\text{disc}}) = \pi_k / \beta$. Normalizing the geometric series $\sum_{k=0}^{k_{\text{max}}} (1/\beta)^k$ yields (3). The drift of K is $\lambda_{\text{intro}} - \lambda_{\text{disc}} = \lambda_{\text{intro}}(1 - \beta)$, positive iff $\beta < 1$. $\square$

The system is reachable iff a live domain exists and some endpoint has a clear address. Because the two layers evolve independently, availability factorizes.

Theorem 1 (Closed-form availability). *The stationary time-average availability of a configuration $(n, \mu, \lambda_a, \beta, k_{\text{max}})$ is*

$$A = \underbrace{(1 - \pi_0(\beta, k_{\text{max}}))}_{\text{domain layer}} \cdot \underbrace{\left(1 - \left(\frac{\lambda_a}{\lambda_a + \mu}\right)^n\right)}_{\text{address layer}}. \quad (4)$$

Proof. Availability is the probability that $K \geq 1$ and at least one of the n addresses is clear. The domain pool $K(t)$

and the n address chains are driven by independent Poisson clocks, so the two events are independent, with probabilities $1 - \pi_0$ (Proposition 1) and $1 - (\lambda_a / (\lambda_a + \mu))^n$ (Lemma 1), respectively. $\square$

Remark 1 (Correlated discovery and timescale separation). The factorization in (4) treats the address and domain layers as independent, whereas a single discovery reveals an endpoint's address and its domain together, so the two channels are correlated in time. The product form nonetheless holds to high accuracy by *timescale separation*. Whenever rotation is cheap, the address layer mixes on the fast timescale of μ and λ_a while the domain pool evolves on the slow timescale of λ_{intro} and λ_{disc} , so the address layer sits in quasi-stationarity at every instant of the domain layer and the factorization is exact in the separation limit. More importantly, the structural results that follow—the impossibility of pure address rotation (Theorem 2) and the phase transition at the frontier—depend only on the drift of the domain pool, not on independence, and so are unaffected by correlation between the two discovery channels.

Equation (4) already exposes the asymmetry. The address factor is driven to 1 by modest redundancy n , whereas the domain factor is controlled entirely by β and the buffer k_{max} . The next result shows that no amount of rotation speed can compensate once $\beta > 1$.

Theorem 2 (Impossibility of pure address rotation). *For $\beta > 1$, regardless of rotation speed μ , endpoint count n , or buffer k_{max} , the time-average availability obeys*

$$A \leq 1 - \pi_0(\beta, k_{\text{max}}) \leq \frac{1}{\beta} < 1, \quad (5)$$

and $1 - \pi_0 \rightarrow 1/\beta$ as $k_{\text{max}} \rightarrow \infty$, so the bound is tight. Availability is therefore bounded away from 1 for any rotation speed, and no (α, T) -availability with $\alpha > 1/\beta$ is achievable. A defender that mints fresh domains more slowly than the censor burns them cannot sustain high availability under a domain-filtering censor. The value $\beta = 1$ is the critical boundary, and only $\beta < 1$ drives the empty-pool probability π_0 , and with it the frontier of §6.1, small.

Proof. The address factor in (4) is at most 1, so $A \leq 1 - \pi_0$. For $\beta > 1$, $(1/\beta)^{k_{\text{max}}+1} \in (0, 1)$ and $\pi_0 = (1 - 1/\beta) / (1 - (1/\beta)^{k_{\text{max}}+1}) \geq 1 - 1/\beta$, whence $1 - \pi_0 \leq 1/\beta < 1$, and letting $k_{\text{max}} \rightarrow \infty$ gives $1 - \pi_0 \rightarrow 1/\beta$. Since $A \leq 1/\beta$ and time-average availability upper bounds the probability that an interval of positive length is fully reachable, (α, T) -availability requires $\alpha \leq 1/\beta$. $\square$

Theorem 2 is the formal content of the claim that rotating faster is the wrong lever. Rotation speed μ appears only in the address factor, which is already saturated by redundancy, and it is absent from the binding domain factor.

What λ_{disc} is, and robustness to its form. Here λ_{disc} is the censor's persistent-block throughput, the rate at which it commits durable blocks into its enforcement path, not its discovery rate. Discovery can be near-instant, since certificate-transparency logs and passive DNS expose issued

names, but committing each block (a blocklist entry, a DPI rule, a poisoned record) is capacity-limited, so treating λ_{disc} as an aggregate the censor runs at full capacity is conservative. The constant-rate *model* is conservative in a second sense: if the censor instead blocks each live unit independently at per-unit rate δ , the aggregate burn δK rises with the pool, adding a stabilizing feedback.

Proposition 2 (State-dependent discovery is favorable). *Under per-unit discovery (aggregate burn δK) the pool is the truncated-Poisson chain $\pi_K \propto (\lambda_{\text{intro}}/\delta)^K/K!$ on $\{0, \dots, k_{\text{max}}\}$, and at a burn matched to the constant-rate model at the full pool ($\delta k_{\text{max}} = \lambda_{\text{disc}}$) it is far more available. At $k_{\text{max}} = 8$, $\beta = 1.6$ the constant-rate pool is empty 38% of the time but the per-unit pool only 0.7%. The constant- λ_{disc} frontier is thus a worst case among discovery models that scale with pool occupancy.*

The genuinely adverse direction is the opposite, discovery that scales with the defender’s *minting* (every issued certificate is logged), which pins β near a constant the defender cannot lower by minting faster. The defenses are then structural rather than rate-based: wildcard certificates that place many names behind one logged entry, and diversification across providers with independent enforcement paths (§8.7). Appendix B derives the chain and Figure 5(c) confirms both directions in simulation.

6.1. The Sustainability Frontier

We now characterize the boundary between sustainable and unsustainable configurations.

Theorem 3 (Sustainability frontier). *Fix $(\alpha, n, \mu, \lambda_a, k_{\text{max}})$ with the address factor $c_a := 1 - (\lambda_a/(\lambda_a + \mu))^n \geq \alpha$. Then time-average availability $A \geq \alpha$ iff $\beta \leq \beta_{\text{avg}}^*$, where the threshold $\beta_{\text{avg}}^*(\alpha, n, \mu, \lambda_a, k_{\text{max}})$ is the unique root of*

$$1 - \pi_0(\beta, k_{\text{max}}) = \frac{\alpha}{c_a}. \quad (6)$$

It is increasing in k_{max} and in n , and $\beta_{\text{avg}}^ \rightarrow c_a/\alpha$ as $k_{\text{max}} \rightarrow \infty$. If $c_a < \alpha$, no β suffices.*

Proof. A in (4) is continuous and strictly decreasing in β (since π_0 is increasing in β), so $A \geq \alpha$ on $(0, \beta_{\text{avg}}^*]$ with β_{avg}^* the solution of $A = \alpha$, i.e. of (6), and existence requires the right-hand side ≤ 1 , i.e. $c_a \geq \alpha$. Monotonicity of π_0 in k_{max} and of c_a in n gives the stated monotonicities. As $k_{\text{max}} \rightarrow \infty$, $1 - \pi_0 \rightarrow 1/\beta$ for $\beta > 1$, so the root of (6) tends to c_a/α . $\square$

The time-average frontier β_{avg}^* thus lies near 1, tending to $c_a/\alpha \in [1, 1/\alpha]$ for a strong address layer. Definition 2 asks for the stricter *interval* guarantee, whose frontier $\beta^* := \beta^*(\alpha, T)$ no closed form captures exactly, since it depends on the joint excursion structure of $K(t)$ and the address chains. We characterize β^* empirically in §8. It satisfies $\beta^* \leq \beta_{\text{avg}}^*$, is below 1 in every configuration we test, and rises toward 1 as the buffer k_{max} grows. The operational conclusion is therefore that *sustained* (α, T) -availability

requires keeping the domain burn rate strictly below a frontier $\beta^ \leq 1$.*

6.2. The Censor’s Best Response

Treating the censor as a Stackelberg leader that allocates a fixed discovery budget B between address discovery (λ_a) and domain discovery (λ_{disc}), the analysis above pins down its optimal play.

Proposition 3 (Optimal censor strategy). *Let the censor split a discovery budget $B = \lambda_a + \lambda_{\text{disc}}$ to minimize availability (4). The denial gain from address discovery, $-\partial A/\partial \lambda_a$, is proportional to $n(\lambda_a/(\lambda_a + \mu))^{n-1}$ times the domain factor and vanishes geometrically in n once redundancy saturates the address factor. The denial gain from domain discovery, $-\partial A/\partial \lambda_{\text{disc}} = (1/\lambda_{\text{intro}}) c_a \partial \pi_0/\partial \beta$, does not saturate while $\beta < 1$. Hence for any n above the redundancy threshold of Corollary 1, the censor’s best response allocates essentially all budget to the domain channel, driving β upward, and the induced game reduces to the domain race of (1).*

This justifies focusing the entire analysis on β . A rational censor, recognizing that address blocking is defeated by redundancy (Lemma 1), concentrates on burning domains, which is what the GFW’s 2024 QUIC-by-domain filtering does in practice [4].

7. The Censor–Defender Simulator

The durable contributions above are independent of any single cloud or censor instance, so they are best tested in a reproducible environment any researcher can rerun. Controlled study of nation-state censors at scale needs privileged access or risky in-country deployment, and real-world confounds (ISP variation, time-of-day, political events) preclude the ablations science requires. We therefore validate in an open, event-driven *censor–defender simulator*.

Design. The simulator is a continuous-time (Gillespie / next-event) Markov simulation of Definition 1. It maintains the domain pool $K(t)$ as a capped birth–death chain and the n address chains (collapsed to a count, since endpoints are exchangeable), and after a warm-up transient it records three quantities. These are (i) the time-average availability, (ii) the (α, T) -interval availability of Definition 2, computed exactly from the simulated up/down excursion structure (Appendix A), and (iii) diagnostics (mean live domains, pool-empty probability, empirical address denial). Each event is scalar arithmetic, so long horizons and many seeds run in well under a second. Because the address and domain layers are driven by independent clocks (Appendix A), the simulator reuses one address realization across an entire domain sweep, which is the bulk of its speed. The full sweeps in this paper complete in a few minutes on one core, with no GPU, cloud, or specialized hardware. Full implementation and the data-fitting calibration procedure are in Appendix E.

What the simulator is and is not. The simulator is a faithful realization of the *model* rather than a network emulator. It does not replay packets, model TCP dynamics, or emulate DPI internals, by design. The durable claims are properties of the timing game, namely the geometric address bound, the birth–death dichotomy, and the phase transition, which are statements about the model and are therefore what a model-level simulator can validate. Mechanism-level questions a model cannot answer (QUIC migration downtime, per-flow detectability, registry latency) are out of scope here and belong to a packet-level testbed. Working at the model level is also what lets the closed-form law and the simulation cross-check each other (Table 2) rather than both depending on the same opaque emulation.

Adversary profiles. Adversary *mechanisms* are fixed to documented designs, and their free rates are expressed in dimensionless form so they transfer to any wall-clock setting. We emphasize that the three profiles in Table 1 are *plausible operating points*, not fits to measurement data: the GFW-like profile has fast active probing and aggressive, collateral-eroded domain filtering [4], [37], the TSPU-like profile has slower per-endpoint discovery [5], and the Iran-like profile follows protocol whitelisting [7]. We do not claim these reproduce any specific censor at a point in time; rather, we report every result as a function of the adversary parameters over a sensitivity sweep, so the qualitative content survives mis-estimation of any one of them, and we configure the profiles *favorably* (faster discovery, larger γ) where uncertain, keeping guarantees conservative. The dimensionless ranges we sweep do bracket published behavior: the GFW acts on a discovered endpoint within minutes [4], [37], Tor bridges are enumerated over days to weeks [13], and the deployed moving-target resolver NinjaDoH held $> 95\%$ availability at 60 s rotation under 30 s discovery [17], all of which lie inside our sweeps. The simulator is designed to be calibrated in the strict sense, its free rates fit to public OONI [65] and Censored Planet [66], [67] archives by replaying historical measurement events, and we leave that data fitting to deployment.

TABLE 1. ADVERSARY PROFILES (DIMENSIONLESS, WITH ROTATION RATE $\mu = 3$).

Censor	address disc. λ_a	μ/λ_a	collateral γ
GFW (China)	1.5	2.00	0.20
TSPU (Russia)	0.8	3.75	0.05
Iran	1.0	3.00	0.10

8. Evaluation

We confirm each analytical result in the simulator. This is model-level validation: the closed-form analysis and an independent simulation agree, and the qualitative predictions survive stresses the closed form omits (state-dependent discovery, bursty and provider-correlated burns); it is not a

fit to a specific censor, which would need the measurement-archive calibration we leave to deployment (§8.13). Unless noted, the canonical configuration is $n = 8$ endpoints, $\mu/\lambda_a = 3$, buffer $k_{\max} = 8$, target $\alpha = 0.95$, averaged over independent seeds with standard-deviation bands.

8.1. Address Rotation Defeats an Address-Only Censor

Figure 1 plots the empirical address-denial probability against the number of endpoints n , for several rotation-to-discovery ratios μ/λ_a , overlaid on the geometric bound of Lemma 1. The simulation matches the closed form to three significant figures, and denial decays geometrically. When discovery is as fast as rotation ($\mu/\lambda_a = 1$), 7 endpoints drive denial below 10^{-2} , and even when discovery is twice as fast as rotation ($\mu/\lambda_a = 0.5$) modest redundancy keeps it small. Address exhaustion is not the binding constraint.

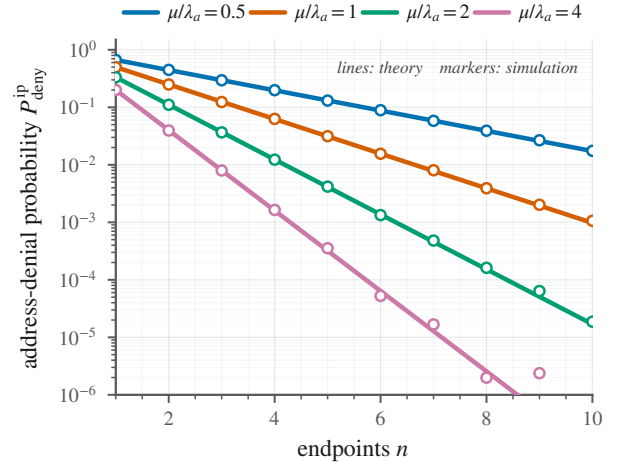


Figure 1. Address blocking is a losing strategy. Denial probability $P_{\text{deny}}^{\text{ip}}$ decays geometrically in the number of endpoints n , and simulation (markers) matches the geometric bound of Lemma 1 (lines). Even when the censor discovers endpoints faster than they rotate ($\mu/\lambda_a < 1$), modest redundancy drives denial to zero.

8.2. The Phase Transition at β^*

Figure 2 is the central result. As the domain burn rate β increases, (α, T) -availability undergoes a sharp phase transition, staying near one while $\beta < \beta^*$ and collapsing past it. Larger domain buffers $k_{\max}$ push the frontier toward $\beta = 1$ and sharpen the transition, as Theorem 3 predicts (β^* increasing in $k_{\max}$, $\beta^* \rightarrow 1$ as $k_{\max} \rightarrow \infty$). The empirical interval frontier lies below the time-average crossing of the closed-form law (4), confirming that the stricter interval guarantee of Definition 2 is the binding one. For the canonical buffer $k_{\max} = 8$ the interval frontier is $\beta^* \approx 0.6$, versus a time-average crossing near $\beta \approx 0.8$.

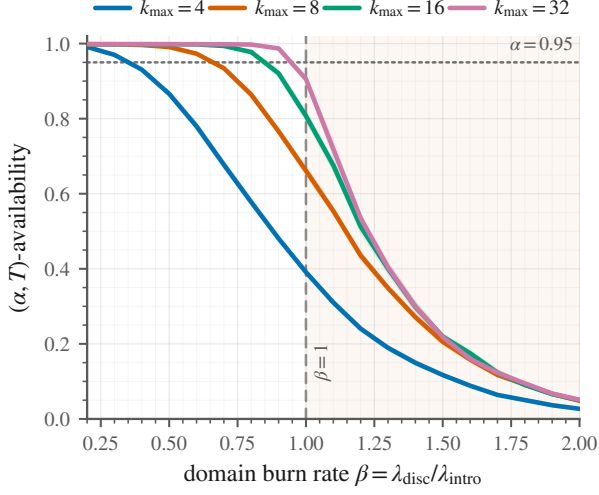


Figure 2. The sustainability frontier. (α, T) -availability vs. the domain burn rate $\beta = \lambda_{\text{disc}}/\lambda_{\text{intro}}$, for domain buffers $k_{\text{max}} \in \{4, 8, 16, 32\}$. Availability collapses once β crosses the frontier $\beta^* \leq 1$, and a larger domain buffer moves β^* toward 1 and sharpens the transition. The dotted line is the (α, T) target and the dashed vertical marks $\beta = 1$.

8.3. The Cost of a Rotation-Only Strategy

Pricing the conventional playbook makes the consequence concrete. A defender that maximizes rotation speed, grows the IP pool, and treats domains as an afterthought operates at, say, $\beta = 1.5$; by Figure 2 its (α, T) -availability is below 0.2 even at the largest buffer and rotation speed we test. A *domain-aware* defender holding $\beta = 0.6$ with the same n , μ , and pool sits above 0.95. The two differ only in the one parameter the rotation-only playbook ignores, the domain-introduction rate, and the failure is hard to detect, since address-layer metrics stay healthy until the censor’s burn rate crosses λ_{intro} , where availability drops sharply with no warning on the rotation side.

8.4. Validation of the Closed-Form Law

Before drawing operational conclusions we check that the simulator and the analysis agree. Table 2 compares the time-average availability predicted by the closed-form law (4) against the simulated value across the full range of β at the canonical buffer $k_{\text{max}} = 8$. The two agree to three significant figures everywhere, confirming both the product-form factorization of Theorem 1 and the correctness of the event-driven simulator. The table also reports the simulated *interval* availability, which is uniformly lower than the time-average value and drops faster as β grows, the quantitative statement that Definition 2’s interval guarantee is the binding one, and the reason the operational frontier β^* sits well below the time-average crossing.

8.5. Rotation Speed Is Not the Lever

Figure 3 sweeps availability over the $(\beta, \mu/\lambda_a)$ plane. Availability is organized almost entirely by β , the horizon-

TABLE 2. CLOSED-FORM LAW (4) VS. SIMULATION ($k_{\text{max}} = 8$, CANONICAL CONFIG). TIME-AVERAGE AVAILABILITY MATCHES TO THREE SIGNIFICANT FIGURES, AND THE INTERVAL METRIC IS THE STRICTER, BINDING ONE.

β	0.4	0.6	0.8	0.9	1.0	1.2	1.5	2.0
closed form	1.00	.993	.961	.930	.889	.793	.658	.499
sim. (avg)	1.00	.993	.961	.928	.890	.791	.658	.498
sim. (intvl)	.996	.973	.864	.766	.661	.435	.206	.048

tal axis. Once rotation is modestly faster than discovery ($\mu/\lambda_a \gtrsim 1$), increasing it further, which moves *up* the figure, does not cross the frontier. No rotation speed restores availability for $\beta \geq 1$. This is the empirical counterpart of Theorem 2, since μ saturates the address layer but is absent from the binding domain layer.

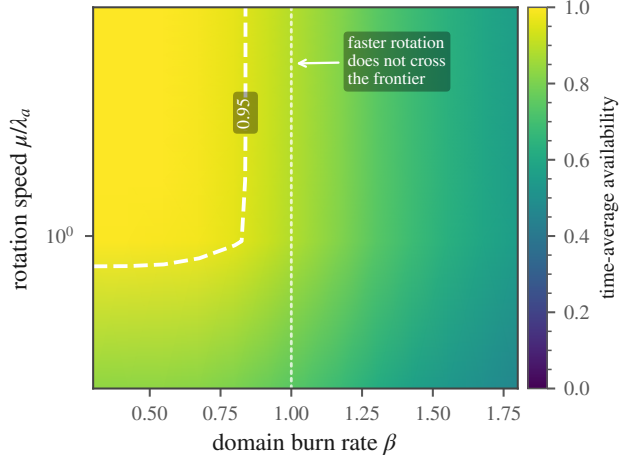


Figure 3. Rotation speed is not the lever. Availability over the $(\beta, \mu/\lambda_a)$ plane, with the white contour at 0.95. The structure is vertical, so crossing the frontier requires lowering β rather than raising rotation speed.

8.6. Longer Sessions Tighten the Frontier

The interval guarantee of Definition 2 depends on the session length T , because a user holding a single connection for longer is more likely to be interrupted by a transient domain-pool depletion. Figure 4a shows (α, T) -availability versus β for session lengths T spanning a factor of twenty, and longer sessions shift the curve down and the frontier left. Figure 4b extracts the interval frontier $\beta^*(\alpha, T)$ over a dense range of T . For $\alpha = 0.95$ it falls from $\beta^* \approx 0.75$ at the shortest sessions to $\beta^* \approx 0.5$ at the longest, and a stricter target $\alpha = 0.99$ lowers it to $\beta^* \approx 0.35$ at the longest sessions. The dependence is smooth and modest, since an order of magnitude in T costs roughly 0.2 in β^* , so a defender sizing its domain economy can pick β from a single design curve once it fixes a target session length and reliability. The renewal approximation of Appendix C predicts these frontiers in closed form and tracks the simulated values to within the grid resolution.

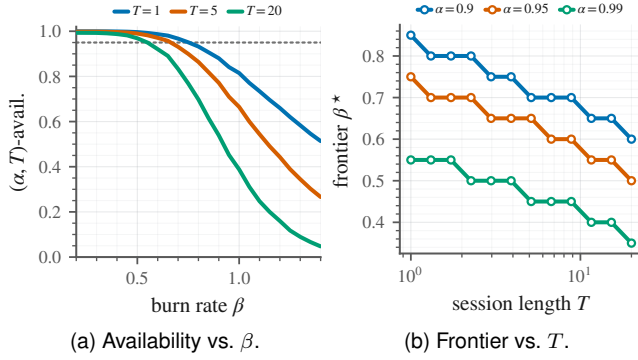


Figure 4. Longer sessions tighten the interval frontier (dotted line $\alpha = 0.95$; T in units of the mean introduction interval). (a) (α, T) -availability vs. β for $T \in \{1, 5, 20\}$; longer sessions shift the curve down and the frontier left. (b) The extracted interval frontier $\beta^*(\alpha, T)$ vs. T (log axis); higher α and longer T both lower β^* , gently.

8.7. Robustness to Bursty Burns

Our model burns domains one at a time, but real domain economies fail in correlated batches. A registrar takedown or a certificate-authority revocation can burn many of a defender’s domains at once. We stress this by having the censor burn b domains per discovery event while *holding the mean burn rate fixed* (events fire $b \times$ less often), so β is unchanged and only the burst structure differs. Figure 5(a) shows that burstiness shifts the frontier sharply left. The interval frontier falls from $\beta^* = 0.6$ at $b = 1$ to 0.5 at $b = 2$ and 0.3 at $b = 4$, and at $b = 8$ no β in the swept range is sustainable. Correlated burns are far more damaging than their mean rate suggests, because a single event can empty the buffer outright. The practical implication is that domain diversity, spreading endpoints across *independent* registrars, authorities, and platforms so that no single takedown is a batch burn, is as important as the raw minting rate. A defender that mints fast but concentrates its domains under one registrar operates at an effective $b \gg 1$ and a correspondingly collapsed frontier.

8.8. Diversification Across Providers Restores the Frontier

The bursty-burn result motivates a concrete defense (design principle P3, §9), namely sourcing domains from several *independent* providers so that one takedown cannot empty the whole pool. We model this directly. The total minting and takedown rates are held fixed and split equally over P providers, and a takedown now empties an *entire* provider’s pool at once, the worst case of full correlation within a provider but independence across them. The system is reachable as long as any one provider still has a live domain. Figure 5(b) shows the payoff. With one or two providers ($P \leq 2$) every takedown is catastrophic and *no* β in the swept range is sustainable, but spreading the same total domain economy across more providers restores the

frontier rapidly, reaching $\beta^* \approx 0.4$ at $P = 4$ and $\beta^* \approx 1.2$ at $P = 8$. With eight independent providers the frontier even exceeds 1, because the probability that *all* providers are simultaneously empty falls off so fast that the defender tolerates a censor burning domains faster than it mints them. This does not contradict the single-pool impossibility of Theorem 2 or the frontier $\beta^* \leq 1$ of §6.1, both of which describe *one* birth–death pool with aggregate burn rate β . Diversification is a different resource model— P parallel pools with union reachability—and is precisely the structural change that lets a defender escape the single-pool bound and remain available at an aggregate $\beta > 1$. Diversification is therefore a quantitative lever comparable to the minting rate itself, available at no extra domain cost, requiring only the operational effort of using multiple registrars and platforms.

Proposition 4 (Diversification law). *If the system draws on P provider pools that fail independently, each empty a fraction $q \in (0, 1)$ of the time, and is reachable whenever any pool is non-empty, then by independence the time-average availability is $A_P = 1 - q^P$. This approaches 1 geometrically in P , and the frontier β_P^* , the largest β with $A_P \geq \alpha$, satisfies $q(\beta_P^*) = (1 - \alpha)^{1/P}$, so it grows with P and exceeds 1 once $(1 - \alpha)^{1/P} > q(1)$.*

Figure 5(d) confirms the simulated A_P matches $1 - q^P$ with the measured per-provider empty fraction q , the theorem-level form of this result. Two further stresses appear in the same figure. Figure 5(c) replaces the constant aggregate burn with *per-unit* discovery (Proposition 2): the interval frontier rises from $\beta^* \approx 0.6$ to $\beta^* \approx 1.67$, since occupancy-scaling discovery makes the pool strictly more available, so the paper’s constant-rate frontier is the conservative case.

8.9. Sensitivity and Adversary Profiles

Figure 6 reports the closed-form time-average frontier β_{avg}^* of Theorem 3 over the (n, k_{max}) grid for each adversary profile (the operational interval frontier β^* of Fig. 2 lies below these values). The qualitative picture is robust. β_{avg}^* increases with redundancy n and buffer k_{max} and saturates near 1. The adversaries differ where the theory predicts, in the redundancy they demand. The GFW’s faster active probing leaves the address factor below α at $n = 2$ (blank cells, where no β suffices), whereas the slower TSPU is already sustainable at $n = 2$. Above $n \geq 4$ the frontier is essentially adversary-independent, since the address layer saturates and only the domain economy binds. No single profile carries the claim.

8.10. The Censor’s Optimal Budget Split

Proposition 3 predicts that a budget-constrained censor concentrates its discovery effort on whichever resource the defender under-provisions, and, once redundancy is adequate, on the domain channel. Figure 7 tests this directly. We give the censor a fixed discovery budget B split by a

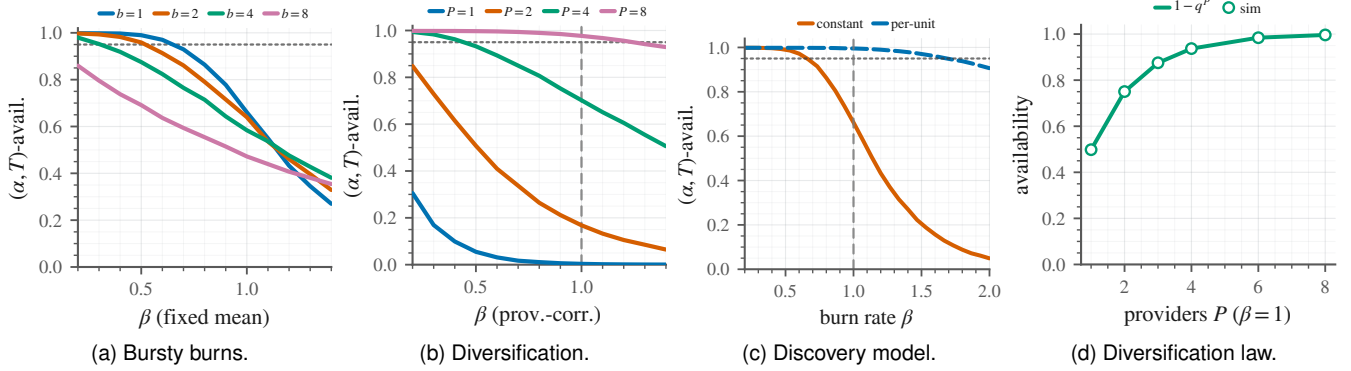


Figure 5. Robustness analyses, showing the frontier is not an artifact of the constant-rate assumption (dotted line $\alpha = 0.95$, dashed vertical $\beta = 1$). **(a)** Bursty burns shift the frontier left as the censor burns b units per event ($\beta^* = 0.6, 0.5, 0.3$ for $b = 1, 2, 4$, none for $b = 8$). **(b)** Diversifying the same economy across P independent providers, under takedowns that empty a whole provider, moves the frontier back past $\beta = 1$. **(c)** Per-unit discovery (aggregate burn δK) is strictly more available than the constant aggregate rate, so the latter is conservative. **(d)** The diversification law: simulated P -provider availability matches the closed form $1 - q^P$.

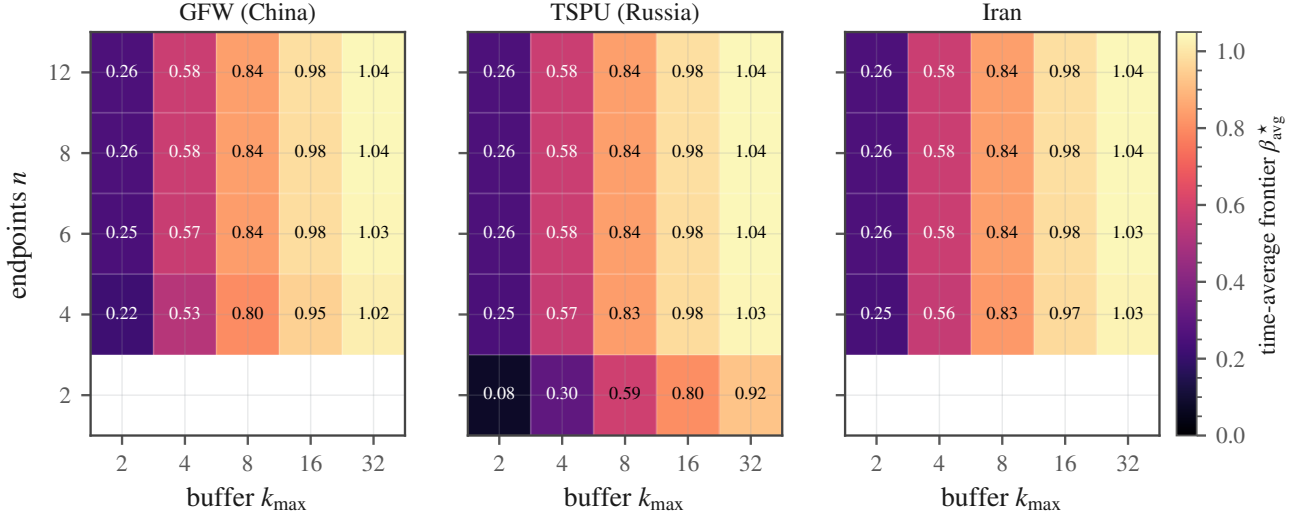


Figure 6. Closed-form time-average frontier β_{avg}^* (Theorem 3) over the (n, k_{max}) grid ($\alpha = 0.95$) under three adversary profiles. Blank cells are configurations whose address factor is below α (no β suffices). β_{avg}^* increases with redundancy n and buffer k_{max} and saturates near 1. The faster GFW censor demands more redundancy (no sustainable β at $n = 2$), and above $n \geq 4$ the frontier is adversary-independent.

fraction f to domain discovery (with the remaining $(1-f)B$ spread over the n endpoints' address discovery) and plot the defender's availability as f varies. The censor's best response is the f that *minimizes* availability. With a single endpoint ($n = 1$) the address layer is the weak point, since availability is lowest near $f \approx 0$, so the censor attacks addresses. Adding even one more endpoint changes the calculus, because for $n \geq 2$ availability is minimized at $f \rightarrow 1$, and the empirical best response is $f^* = 0.97$, that is, essentially all budget to domains. The address layer's geometric defense (Lemma 1) thus does not merely make address blocking ineffective. It makes domain blocking the censor's *only* rational move, as Proposition 3 claims and as the GFW's pivot to domain filtering reflects.

8.11. The Domain-Economy Operating Recipe

The frontier turns directly into an operating recipe. To stay available against a censor that burns domains at rate λ_{disc} , a defender must mint fresh domains at rate $\lambda_{\text{intro}} \geq \lambda_{\text{disc}}/\beta^*$. Figure 8 plots this minimum mint rate against the censor's burn rate, with a secondary cost axis. The relationship is linear with slope $1/\beta^*$, and the absolute rates are modest. Even against a censor blocking hundreds of the defender's units per day, holding $\beta < \beta^*$ requires minting on the order of a few hundred fresh units per day. What that costs is better measured in provider rate limits than in dollars (§6). The cash outlay is small: free-tier platform sub-domains and short-lived ACME certificates are free, and a self-controlled registrable domain costs a few dollars a year while a single *wildcard* certificate covers all

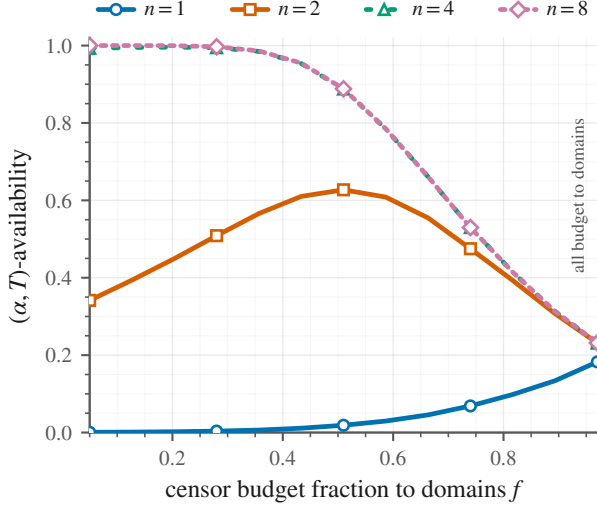


Figure 7. The censor’s optimal budget split. Defender availability vs. the fraction f of a fixed censor discovery budget spent on domains (rest on addresses). The censor minimizes availability. At $n = 1$ it attacks the address layer ($f \rightarrow 0$), but for $n \geq 2$ the minimum is at $f \rightarrow 1$, all budget to domains, validating Proposition 3.

of its sub-domains, so certificates are issued per registrable domain, not per name. At $\beta^* \approx 0.6$ against a censor burning 200 units per day the defender mints roughly 330 fresh names per day, which with wildcard certificates is a handful of certificate orders per day, within free ACME rate limits, for a few dollars per day or less. The binding ceiling under more aggressive burn is not money but the providers’ account-creation and abuse-review rate limits, a further reason to diversify, since P independent providers supply P independent rate-limit budgets (§8.8); an operator on a single registrar or platform hits that ceiling, not a financial one, first. The recipe also makes the censor’s cost explicit. To push a defender off the frontier the censor must sustain a domain-discovery throughput exceeding $\beta^* \lambda_{\text{intro}}$ indefinitely, since blocks are persistent but the defender’s minting is continuous. The contest is thus an economic one between two provisioning rates, and Appendix D shows the defender’s unavoidable cost is set by the censor’s burn rate rather than by how large a buffer it keeps.

8.12. Relation to Deployed Moving-Target DNS

NinjaDoH [17] reported $> 95\%$ availability at 60 s rotation under 30 s discovery, costing \$0.50–2.00/day. Our theory explains why: with discovery roughly twice as fast as rotation, the address factor of (4) needs only modest redundancy (Corollary 1) to exceed 0.95, and the reported cost matches the regime $\beta < \beta^*$. What that work lacked, and this paper supplies, is the characterization of *when* rotation wins and the warning that the same system fails without warning once the domain-burn rate overtakes the minting rate.

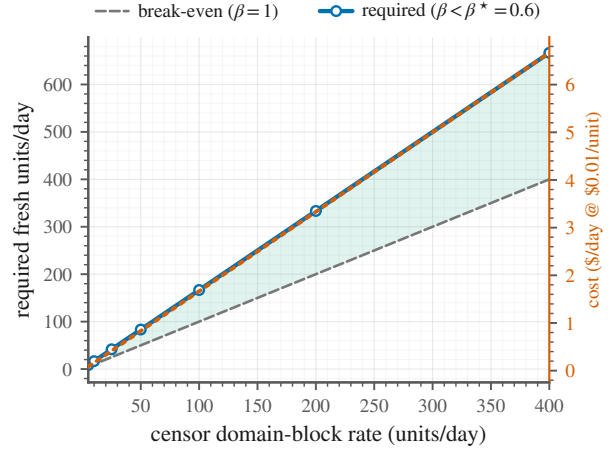


Figure 8. The domain-economy operating recipe. Minimum fresh-domain mint rate needed to hold $\beta < \beta^*$ vs. the censor’s domain-block rate (with a secondary cost axis). The requirement is linear in the censor’s burn rate with slope $1/\beta^*$, so staying available is a question of out-minting rather than out-rotating.

8.13. Threats to Validity

A simulation-primary methodology has one dominant risk, the gap between the model and reality, and we bound it rather than ignore it. *First*, no result rests on a single operating point, and every claim is reported as a function of the adversary parameters ($\gamma, \lambda_a, \lambda_{\text{disc}}$) over a sensitivity sweep (Figs. 2–6), so qualitative content such as the phase transition at β^* survives mis-estimation of any one parameter, and the adversary profiles (Table 1) merely locate plausible operating regions within those sweeps. *Second*, where the censor’s true capabilities are uncertain we configure it favorably, with faster discovery and a larger collateral budget, so the reported frontiers are conservative lower bounds on what a real defender can achieve. *Third*, the qualitative content is structural rather than parametric. The geometric address bound (Lemma 1) and the birth–death dichotomy (Proposition 1) follow from the *drift* of the chains rather than the exact rate distributions, and Appendix B shows the address bound is distribution-free. *Fourth*, the closed-form law and the independent simulator agree to three significant figures (Table 2), so a coding error in one would have to be matched by an analytical error in the other to pass undetected. Beyond these internal checks, the model’s central qualitative prediction, that a rational censor concentrates on domain blocking (Proposition 3), is corroborated by the GFW’s actual 2024 pivot to QUIC-by-domain filtering [4], an external data point the simulation does not supply. The one quantity this methodology cannot self-validate is reachability from *inside* a censored network, that is, whether a hyperscaler IP that our model deems reachable truly carries traffic past a real middlebox. That is an empirical question orthogonal to the frontier, since it shifts the absolute timescale rather than the existence or location of β^* , and is best answered by passive remote measurement, which we leave to future work.

9. Design Implications and Operating Blueprint

The theory is not only diagnostic; it prescribes how to build and operate a moving-target service, in four principles. **P1 (move, don’t hide):** the censor is assumed to find every endpoint, so security comes from rotation outpacing action; the address layer is won with a handful of endpoints (Lemma 1), and the complexity budget belongs on the domain economy. **P2 (make minting, not rotation, the first-class control):** few systems expose a domain-introduction rate, yet Theorem 2 says it is what decides availability, so a platform should instrument λ_{intro} and the live-unit count K and alarm on β approaching β^* . **P3 (diversify across independent providers):** correlated burns collapse the frontier (§8.7), so units should be spread across independent registrars, authorities, and platforms, since concentration under one yields an effective $b \gg 1$. **P4 (size the buffer for resilience, not cost):** the buffer k_{max} raises the frontier and absorbs bursts, but the unavoidable cost is censor-set ($c_{\text{mint}}\alpha\lambda_{\text{disc}}$, independent of k_{max} ; Appendix D), so k_{max} should track the session length T and the expected burst size.

Estimating β from telemetry. Both rates in $\beta = \lambda_{\text{disc}}/\lambda_{\text{intro}}$ are observable to the operator. It sets λ_{intro} directly, and measures λ_{disc} from the channel-health monitor it already runs: the empirical burn rate is the count of live units that go from reachable to persistently unreachable per unit time (probed from passive remote vantage points, as in §8). Their ratio is β , refreshed online and alarmed on as it nears β^* ; the burst size b (§8.7) is read off the same trace, and λ_a, μ are the operator’s own telemetry. No quantity requires privileged access to the censor.

An operating recipe. To be (α, T) -available against a censor with domain-burn rate λ_{disc} , a defender takes five steps. It runs $n \geq \lceil \ln(1/(1-\alpha))/\ln((\lambda_a + \mu)/\lambda_a) \rceil$ endpoints (Corollary 1), reads the frontier $\beta^*(\alpha, T)$ off Fig. 4b for the chosen T , mints at $\lambda_{\text{intro}} \geq \lambda_{\text{disc}}/\beta^*$ (Fig. 8), holds a buffer k_{max} at least the expected burst size, and sources units from $\geq b$ independent providers. The whole recipe is computable from quantities a defender can measure, and none of it requires out-rotating the censor.

10. Discussion

Relation to the anonymity trilemma. Lower bounds are known for *anonymity* (the trilemma trades anonymity, bandwidth, and latency) but not, previously, for moving-target availability. Our frontier is an availability analogue, except that the binding resource is economic (blockable units) rather than informational, so collateral freedom and the domain economy, not entropy, set the limit.

What the censor can do about it. The model maps each rational counter-move to a parameter. Raising domain-discovery throughput λ_{disc} increases β but is answered by

raising λ_{intro} , a race the cheaper-unit side wins; eroding the collateral budget γ attacks the over-provisioned address layer and buys little until $\gamma \rightarrow 0$, where the premise itself fails; and correlated burns (§8.7), the most cost-effective move, are answered by diversification. In every case the model turns “the censor will adapt” into a specific knob and a specific defense.

Generality beyond DNS and proxies. The game abstracts away the service: the resource is an (address, unit) pair and the payoff is reachability, so the frontier applies to *any* service on rotating cloud infrastructure (a news mirror, a messaging entry point, a human-rights API), with only the wall-clock calibration of the rates changing. The dimensionless frontier β^* and the impossibility for $\beta > 1$ are invariant, which is why we frame the contribution as a theory rather than a system.

From rotation to the full stack. This paper characterizes the rotation layer. The same timing-game lens extends to the two other layers a complete system needs, an enumeration-resistant discovery directory and a traffic-shaping layer with provable indistinguishability, and to a unifying controller over both, none of which our reachability analysis addresses. The rotation theory here is the foundation those layers build on.

Ethical considerations. This work is theory and simulation, and it involves no human subjects, no in-country deployment, and no user data. The simulator encodes only already-published censor capabilities [2], [4], [5], conferring no advantage a nation-state lacks while giving defenders a shared testbed, an asymmetry that favors openness. We release all artifacts under permissive licenses. Appendix F gives a fuller ethics discussion.

11. Conclusion

We have given, to our knowledge, the first formal account of *when* moving-target censorship resistance works. Modeling censor–defender interaction as a timing game on a combinatorial (address, domain) space, we proved that the address resource is free (address blocking decays geometrically in redundancy) and the domain resource is binding, derived a closed-form availability law, and proved a sustainability frontier. With a single domain pool, availability is sustainable iff the domain burn rate $\beta = \lambda_{\text{disc}}/\lambda_{\text{intro}}$ stays below a threshold $\beta^* \leq 1$, and no rotation speed can compensate once $\beta > 1$; diversifying across independent providers raises the frontier and can push it past 1. An open simulator reproduces the predicted phase transition at β^* across adversary profiles representative of the GFW, TSPU, and Iran, and turns the theory into a concrete operating recipe. The result reframes a decade of system design. The right knob is not rotating IPs faster but keeping the domain economy ahead of the censor, and the theory gives operators a falsifiable health metric, the domain burn rate β , and a threshold to stay below, packaged with an open simulator any researcher can rerun.

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Appendix A. Detailed Derivations

This appendix collects the technical details deferred from §5–6.1.

Product-form stationarity. The full state is (c, K) , where $c \in \{0, \dots, n\}$ is the number of clear addresses and $K \in \{0, \dots, k_{\max}\}$ the number of live unblocked domains. Both coordinates are driven by independent Poisson clocks. The count c increases at rate $\mu(n - c)$ (rotations restoring a blocked address) and decreases at rate $\lambda_a c$ (discoveries), while K increases at rate $\lambda_{\text{intro}} \mathbf{1}[K < k_{\max}]$ and decreases

at rate $\lambda_{\text{disc}} \mathbf{1}[K \geq 1]$. The generator is therefore the Kronecker sum of the two one-dimensional generators, so the stationary distribution factorizes, $\pi(c, K) = \pi_c \pi_K$, with π_c binomial $\text{Bin}(n, \mu/(\mu + \lambda_a))$ and $\pi_K \propto (1/\beta)^K$. Reachability is the event $\{c \geq 1\} \cap \{K \geq 1\}$, whose stationary probability is the product $(1 - \pi_0^c)(1 - \pi_0^K)$, which is (4). This is the formal basis for treating the two layers separately throughout.

Exactness of the interval metric. Let $\{[s_j, e_j]\}$ be the maximal intervals on $[0, H]$ during which the system is unreachable (the union of the address-down and domain-down excursions). A window $[t_0, t_0 + T]$ is fully reachable iff it meets no $[s_j, e_j]$, i.e. iff $t_0 \notin \bigcup_j (s_j - T, e_j)$. Hence the exact (α, T) -interval availability over the run is

$$1 - \frac{|\bigcup_j (s_j - T, e_j) \cap [0, H - T]|}{H - T}, \quad (7)$$

which the simulator computes by merging the (at most) $|\{j\}|$ widened intervals in $O(J \log J)$ time. No windowing approximation is involved.

Appendix B.

General (Non-Poisson) Discovery

Lemma 1 assumed exponential discovery. We show the geometric product bound is distribution-free.

Lemma 2 (General-discovery address bound). *Suppose each endpoint rotates at the points of a rate- μ Poisson process and, after each rotation, is discovered after an independent delay $D \sim G$ with mean $\mathbb{E}[D] < \infty$. Then the stationary probability that the endpoint's address is blocked is $p = \mu \mathbb{E}[(I - D)^+]$, where $I \sim \text{Exp}(\mu)$ is an inter-rotation interval independent of D , and the n -endpoint denial probability is $P_{\text{deny}}^{\text{ip}} \leq p^n$. For $D \sim \text{Exp}(\lambda_a)$, $p = \lambda_a/(\lambda_a + \mu)$, recovering Lemma 1.*

Proof. By renewal-reward over an inter-rotation cycle of mean length $\mathbb{E}[I] = 1/\mu$, the fraction of time blocked is $\mathbb{E}[(I - D)^+]/\mathbb{E}[I] = \mu \mathbb{E}[(I - D)^+]$. Independence across endpoints (each draws a fresh address on rotation) gives the product bound, with strict inequality when the collateral cap binds. For $D \sim \text{Exp}(\lambda_a)$, $\mathbb{E}[(I - D)^+] = \int_0^\infty \lambda_a e^{-\lambda_a d} \frac{1}{\mu} e^{-\mu d} dd = \lambda_a/(\mu(\lambda_a + \mu))$, so $p = \lambda_a/(\lambda_a + \mu)$. $\square$

Remark 2. Among discovery laws with a fixed mean, heavy-tailed D (occasional very slow discovery) lowers $\mathbb{E}[(I - D)^+]$ and hence p , so erratic censors help the defender. The exponential case is a convenient, near-worst middle ground, so using it keeps the address-layer guarantees conservative.

State-dependent domain discovery (Proposition 2). The domain analogue relaxes the constant burn rate of Proposition 1. If the censor blocks each of the K live units independently at per-unit rate δ , the pool is a birth-death chain with birth λ_{intro} (while $K < k_{\text{max}}$) and death δK . Detailed balance gives $\pi_{K+1} = \pi_K \lambda_{\text{intro}}/(\delta(K+1))$, hence

$\pi_K \propto (\lambda_{\text{intro}}/\delta)^K/K!$, a Poisson law of mean $\rho = \lambda_{\text{intro}}/\delta$ truncated to $\{0, \dots, k_{\text{max}}\}$ with $\pi_0 = (\sum_{j=0}^{k_{\text{max}}} \rho^j/j!)^{-1}$. Matching the constant-rate model at the full pool ($\delta k_{\text{max}} = \lambda_{\text{disc}}$, so $\rho = k_{\text{max}}/\beta$) and comparing with the geometric π_0 of Proposition 1 gives, at $k_{\text{max}} = 8$ and $\beta = 1.6$, $\pi_0 \approx 0.38$ (constant) versus $\pi_0 \approx 7 \times 10^{-3}$ (per-unit). The rising death rate is a restoring force that holds the pool near ρ , so occupancy-scaling discovery is strictly more favorable and the constant-rate frontier is conservative.

Appendix C.

A Renewal Approximation for the Interval Frontier

The time-average frontier β_{avg}^* has a closed form (Theorem 3). The operational interval frontier $\beta^*(\alpha, T)$ does not, but admits an accurate approximation. With a strong address layer the down events are domain-pool depletions. The pool is empty a fraction π_0 of the time, and each empty sojourn lasts $\text{Exp}(\lambda_{\text{intro}})$ in expectation (only a mint escapes state $K = 0$), so depletions begin at rate $\nu = \pi_0 \lambda_{\text{intro}}$. Treating onsets as approximately Poisson,

$$\text{interval-avail}(\beta, k_{\text{max}}, T) \approx (1 - \pi_0) e^{-\pi_0 \lambda_{\text{intro}} T}, \quad (8)$$

and $\beta^*(\alpha, T)$ is the root in β of the right-hand side equalling α . Table 3 compares this prediction against the simulator (canonical config, $k_{\text{max}} = 8$, $\lambda_{\text{intro}} = 1$). The approximation tracks the measured frontier to within about 0.03, the order of the sweep resolution, slightly under-estimating β^* because it treats depletion onsets as independent and so ignores their clustering. Equation (8) is thus a usable design formula, since a defender can read β^* off it without simulation.

TABLE 3. INTERVAL FRONTIER $\beta^*(\alpha=0.95, T)$, RENEWAL APPROXIMATION (8) VS. SIMULATION (§8, $k_{\text{max}} = 8$).

session length T	1	5	20
approximation (8)	0.74	0.62	0.52
simulation	0.75	0.65	0.50

Appendix D.

The Cost-Optimal Domain Economy

Let c_{mint} be the cost of minting one registrable domain and c_{hold} the per-unit-time cost of holding one live domain.

Proposition 5 (Censor-set cost floor). *In steady state the realized minting throughput equals the burn throughput, $\lambda_{\text{disc}} \Pr[K \geq 1]$, so at an (α, T) -available operating point the minting cost rate is*

$$c_{\text{mint}} \lambda_{\text{disc}} \Pr[K \geq 1] \approx c_{\text{mint}} \alpha \lambda_{\text{disc}}, \quad (9)$$

independent of the buffer k_{max} . The total operating cost rate is $c_{\text{mint}} \alpha \lambda_{\text{disc}} + c_{\text{hold}} \mathbb{E}[K]$.

Proof. For the capped birth–death chain, in stationarity the up-flux across each edge equals the down-flux, so the rate of births (mints) equals the rate of deaths (burns), $\lambda_{\text{disc}} \Pr[K \geq 1]$. At an (α, T) -available point $\Pr[K \geq 1] \gtrsim \alpha$. Holding cost is $c_{\text{hold}} \mathbb{E}[K]$ by definition. $\square$

There are two consequences. (i) The *flow* cost of domains is dictated by the censor’s burn rate rather than by how large a buffer the defender keeps, since stockpiling does not raise the bill the censor imposes. (ii) The buffer is therefore a pure resilience knob, and it should be sized to the target session length (Fig. 4b) and expected burst size (§8.7), with holding cost $c_{\text{hold}} \mathbb{E}[K]$ the only penalty. Separately, the required minting *capacity* (peak rate) is $\lambda_{\text{intro}} \geq \lambda_{\text{disc}}/\beta^*$, a provisioning constraint distinct from the per-domain flow cost.

Appendix E. Simulator and Calibration Details

Transition rates. The simulator advances the two layers by the next-event method with the rates of Table 4, and bursty burns (§8.7) replace the domain death by a batch of b removals at rate λ_{disc}/b , leaving the mean burn rate unchanged. Because the layers are independent, one address realization is reused across an entire domain sweep, which is the bulk of the speedup.

TABLE 4. SIMULATOR TRANSITION RATES (NEXT-EVENT / GILLESPIE).

layer	transition	rate
address	$c \rightarrow c - 1$ (block)	$\lambda_a c$
address	$c \rightarrow c + 1$ (rotate/recover)	$\mu(n - c)$
domain	$K \rightarrow K + 1$ (mint), $K < k_{\text{max}}$	λ_{intro}
domain	$K \rightarrow K - 1$ (burn), $K \geq 1$	λ_{disc}

Next-event loop and metric. Each realization advances time by $\text{Exp}(R)$ for total rate R , fires one transition with probability proportional to its rate, and accumulates occupancy. After the warm-up, the unreachable intervals are the union of the address-down ($c = 0$) and domain-down ($K = 0$) excursions, and (α, T) -interval availability is computed exactly from them by the measure of Appendix A. Since the layers share no state, a sweep reuses one address realization per seed across all domain runs, the source of the few-minute runtime.

Calibration methodology. Adversary *mechanisms* are fixed to documented GFW, TSPU, and Iranian designs [2], [4], [5], [7]. In deployment their free rates are fit to public OONI [65] and Censored Planet [66] archives by replaying historical measurement events and maximizing agreement on a held-out split. The results here use the dimensionless profiles of Table 1, chosen adversary-favorably so the guarantees are conservative.

Canonical parameters and reproducibility. Unless noted, the canonical configuration is $n = 8$, $\mu/\lambda_a = 3$, $k_{\text{max}} = 8$, $T = 5$, $\lambda_{\text{intro}} = 1$, and experiments sweep one axis at a time ($\beta \in [0.2, 2.0]$, $\mu/\lambda_a \in [0.25, 8]$, $k_{\text{max}} \in [2, 32]$, $T \in [1, 20]$, $b \in [1, 8]$, $P \in [1, 8]$), each point averaging ≥ 12 seeds with standard-deviation bands over horizons of $1.5\text{--}4 \times 10^4$ units (10% warm-up). The closed-form law (4) matches the simulator to three significant figures across the sweep (Table 2), and one command regenerates every figure. All code, configurations, and results are available under permissive licenses at:

<https://github.com/SecretLabOU/BlockAMole>

Appendix F.

Ethics Considerations. This work consists of theory and discrete-event simulation. It involves no human subjects, no in-country deployment, no live or third-party systems, and no user or network data of any kind. All quantities reported come from a self-contained simulator that any reader can run on a laptop. We followed the principles of beneficence, respect for persons, and justice throughout, and we judge the overall benefit of placing moving-target censorship resistance on provable foundations to outweigh the limited and well-mitigated risks.

Vulnerability disclosure. Not applicable. The work identifies no vulnerability in any deployed system and contains no CVE identifiers. For any defect later found in our released artifacts we will follow coordinated disclosure.

Human subjects research. Not applicable. There are no human subjects and no IRB review applies, and the paper handles no personally identifiable or sensitive data. The only external data are published, aggregate measurement archives (for example OONI and Censored Planet) cited for calibration.

Working with live systems. Not applicable. Nothing outside our own simulator was touched, so no permission was needed and there is no risk of harm or disruption to third-party systems.

Developing new tools or technologies. The theory and open simulator are dual-use. The clearest misuse is that our central finding, that a rational censor should target domain discovery over IP blocking, could steer a censor’s investment. We judge this risk low and well mitigated: the censor has already drawn that conclusion in practice (the GFW’s 2024 QUIC-by-domain filtering), so the guidance confers no advantage a nation-state lacks; the simulator encodes only published capabilities and contains no exploit, evasion payload, or deanonymization method, giving reachability guarantees for legitimate operators rather than an attack tool; and the same artifact materially helps defenders, who had no shared way to reason about when rotation wins. We will follow coordinated disclosure for any implementation issue in released artifacts. On balance the work strengthens free and open access to information far more than it could aid its suppression.